0109026/27/2&1 B.B.E.K.O
September 2026
SCIENCE 2&1
Essay and Objective
2 hours

2 &1

Name.....

Index Number.....

BEST BRAIN EXAMINATIONS KONSORTIUM
GHANA

Special Mock Examinations for BECE Candidates

September 2026

SCIENCE 2 & 1

ESSAY AND OBJECTIVE

2 hours 10 minutes

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in* **ink** *in the spaces provided above.*

This booklet consists of **two** *papers. Answer Paper* **2** *which comes first, in your answer booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last* **1** *hour* **25** minutes *after which the answer booklet will be collected. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last* **45** *minutes.*

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PAPER 2

ESSAY

[100 marks]

.....Name.....

.....Index Number.....

Answer **all** *the questions in your answer booklet*

This paper is in **two** *sections:* **A** *and* **B**. *Answer Question* ~~**1 in section A and any other**~~ **three** *questions in section* **B**.

Credit will be given for clarity of expression and orderly presentation of material.

SECTION A

[40 marks]

Answer all the questions in this section

1. (a) Figure 1(a) is an illustration of some components in an electronic circuit.
Study them carefully and answer the questions that follow.

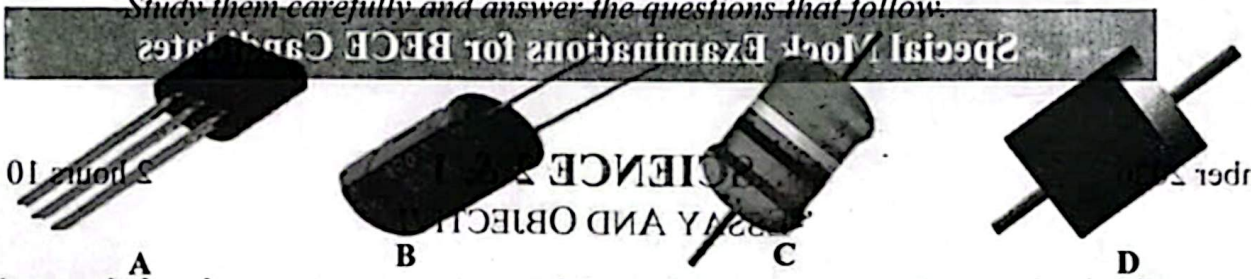


Figure 1(*a*)

(i) Identify **each** of the circuit components labelled **A, B, C** and **D** above.

(ii) Draw the circuit symbol of each of the components labelled **A, B, C** and **D**.

(iii) State **one** function of each of the components labelled **A, B, C** and **D**. [10 marks]

(b) Figure 1(b) is an illustration of the digestive system in the human body.
Study the figure carefully and answer the questions that follow.

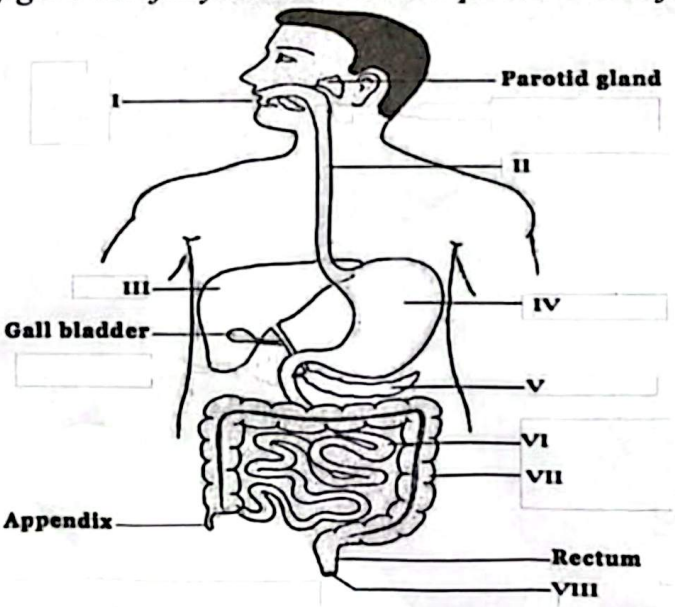


Figure 1(*b*)

(i) Name the parts labelled **I, II, III, IV, V, VI, VII** and **VIII.**

(ii) Describe two functions of the part labelled **I** in the digestion process.

[10 marks]

(c) Figure 1(c) shows the periodic table with the first twenty elements.
Study it carefully and answer the questions that follow.

Figure 1*(c)*

Periods																	
	H																He
	Li	I									B	II	N	O	F		III
	Na	Mg									Al	Si	IV	S	Cl		V
	VI	Ca															

(i) Name the missing elements labelled **I, II, III, IV, V** and **VI.**
(ii) List **two** elements **each** of *Group 2* and *Period 3.*
(iii) Identify **two** elements that are *noble gases.*
(iv) Classify all the elements as metals and non-metals.

[10 marks]

(d) In Figure 1(d) is an illustration of different types of animals reared on a farm.
Study them carefully and answer the questions that follow.

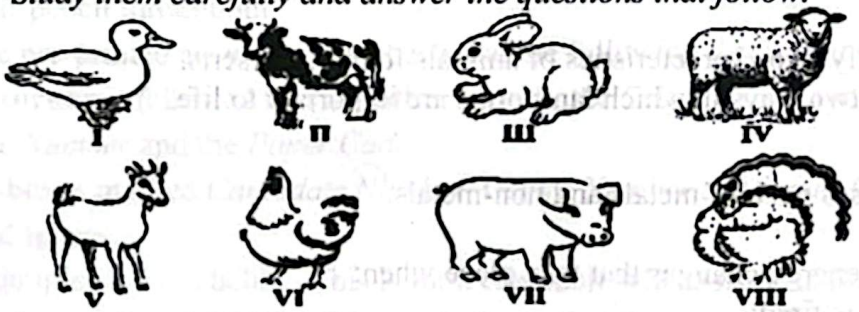


Figure. 1*(d)*

(i) Identify each of the animals labelled **I, II, III, IV, V, VI, VII** and **VIII** above.
(ii) Classify each animal above as ruminant, poultry or monogastric.
(iii) Distinguish between a *monogastric* and a *ruminant.*

[10 marks]

SECTION B
[60 marks]
Answer **three** *questions* **only** *from this section.*

2. (a) Describe **two** ways in which the stomach helps in digestion. [4 marks]

(b) (i) Give **three** uses of poultry.
(ii) State **two** characteristics of ruminants.

[6 marks]

(c) Use the stages in the water cycle in figure 2 to answer the following questions.

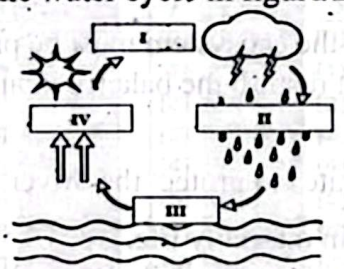


Figure 2

(i) Name and explain each of the stages labeled **I, II, III** and **IV.**
(ii) Give **two** activities that can affect the water cycle.

[6 marks]

Turn over

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d) Give **two** examples of conduction in everyday life. [4 marks]

3. (a) Draw **three** diagrams to show how particles are arranged in **each** of the three states of matter. [6 marks]

(b) (i) State Newton's First Law of Motion.
(ii) If a force of 60N is exerted to accelerate a metallic ball at $3m/s^2$, calculate the mass of the metallic ball.

[4 marks]

(c) (i) Give **two** safety precautions to consider when using a hoe.
(ii) Outline **four** proper ways of handling agricultural tools. [6 marks]

(d) (i) Identify **four** breeding places of houseflies.
(ii) Name **two** diseases transmitted by the housefly. [4 marks]

4. (a) (i) Identify **two** characteristics of animals found in deserts.
(ii) State **two** ways in which landforms are important to life. [4 marks]

(b) State **three** uses **each** of metals and non-metals. [6 marks]

(c) Describe the energy changes that take place when:
(i) a gun is fired;
(ii) a hydropower dam is in operation;
(iii) the palms are rubbed together. [6 marks]

(d) Explain the difference between the following with **one** example **each**;
(i) biodegradable and non-biodegradable wastes.
(ii) hazardous and non-hazardous wastes. [4 marks]

5. (a) Explain the difference between each of the following farming systems;
(i) land rotation and shifting cultivation;
(ii) monoculture and mono cropping;
(iii) mixed cropping and mixed farming. [6 marks]

(b) (i) State **three** reasons why the ecosystem must be preserved.
(ii) Give **three** activities that disturb the balance in an ecosystem. [6 marks]

(c) Explain how a car seat belt operates to protect the driver. [3 marks]

(d) (i) State **two** uses of solids in our daily life.
(ii) Identify **three** useful liquids in our daily life and give the use of each. [5 marks]

END OF ESSAY TEST

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Answer all the questions.

Each question is followed by **four** *options lettered* **A** *to* **D**. *Find the* **correct** *option for* **each** *question and shade* **in pencil** *on your answer sheet, the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

Which of the following elements reacts with water?

A. Carbon
B. Sodium
C. Sulphur
D. Iodine

The correct answer is **Sodium**, *which is lettered* **B** *and therefore answer space* **B** *would be shaded.*

[A] [B] [C] [D]

Think carefully before you shade the spaces; erase completely **any** *answer you wish to change.*

Do **all** *rough work on this question paper.*

Now answer the following questions.

1. Proper waste management is necessary in a community to

I. reduce activities causing pollution.
II. decrease the volume of waste in the community.
III. encourage the creation of more landfills.
IV. discourage recycling of waste.

A. I, II, III, and IV
B. I and II only
C. I, II and III only
D. II, III, and IV only

2. Which of the following is **not** a way of caring for agricultural tools for a longer life span?

A. Regularly cleaning tools to remove dirt.
B. Spraying handles with insecticide to keep flies away.
C. Sharpening the cutting edges regularly.
D. Applying lubricants to prevent rust.

3. The seed drill is a tool used on farms to

A. create deep furrows for planting.
B. compact soil for better drainage.
C. remove debris from soil.
D. create rows for the right seed depth.

4. Acute respiratory syndrome and kidney failure are commonly associated with

A. Ebola.
B. Hepatitis.
C. COVID.
D. Meningitis.

5. How is potential energy converted into kinetic energy in a hydropower dam?

A. Water in the dam rushes through the turbines.
B. Gravity causes water in the reservoir to flow through an inlet in the dam.
C. The turbine turns round on its own at a very high speed.
D. The power plant is powered by the generator to release energy.

6. Monogastric animals are generally known to be

A. having complex stomachs.
B. usually herbivores.
C. herbivores, carnivores or omnivores.
D. usually carnivores.

7. In which type of landform in Ghana are Baobab trees and guinea grass commonly found?

A. Mountainous regions
B. Coastal plains
C. Savanna plateaus
D. River valleys

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8. The first step in the digestive process is
   A. chewing the food.
   B. swallowing the food.
   C. absorbing the nutrients in a food.
   D. eliminating waste.

9. Which of the following is a non-metal found in the first group of the periodic table?
   A. Beryllium
   B. Helium
   C. Lithium
   D. Hydrogen

10. All the following are inner planets except
   A. Neptune.
   B. Earth.
   C. Mercury.
   D. Mars.

11. The main end product of digestion of the starch in kenkey is
   A. amino acid.
   B. glucose.
   C. fatty acids.
   D. glycerol.

12. The component that opposes fluctuations in an electronic circuit is the
   A. resistor.
   B. capacitor.
   C. inductor.
   D. transistor.

13. Which of the following human activities interferes with the carbon cycle?
   A. Destruction of the forest
   B. Reduction in carbon emissions.
   C. Replanting of trees felled as timber
   D. Protection of the forest ecosystem.

14. Which of the following animal breeds is associated with rabbits?
   A. African Dwarf
   B. Ashanti Dwarf
   C. Djallonke
   D. New Zealand White

15. Silicon and boron are elements that are
   A. metals.
   B. non-metals.
   C. semi-metals.
   D. noble gases.

16. Which of the following are true of Jupiter?
   I. It is the largest in the solar system.
   II. It orbits the Sun in 12 Earth days.
   III. It spins very fast.

   A. I and II only
   B. II and III only
   C. I and III only
   D. I, II and III

17. Plants and animals are interdependent in an ecosystem because
   I. Animals feed on plants.
   II. Plants provide oxygen to animals.
   III. Animals get shelter from plants.
   IV. Carnivorous animals feed on animals that depend on plants for food.

   A. I, II, III, and IV
   B. I, II and IV only
   C. I, II and III only
   D. II, III, and IV only

18. Which viral disease is traced to swine and avian genes?
   A. H1N1
   B. N1H1
   C. Newcastle
   D. mumps

19. The effort in a lever is the point where
   A. load is lifted.
   B. the lever turns.
   C. force is applied.
   D. the lever balances.

20. The process of energy transfer from plants to animals in an ecosystem is called a
   A. Water cycle.
   B. Food chain.
   C. Carbon cycle.
   D. Decomposition.

21. The idea of a green economy does **not**
   A. make sustainability a priority.
   B. promote environmental protection.
   C. support depletion of resources.
   D. ensure usage of natural resources.

22. Which of the following agricultural tools is best suited for levelling soil?
   A. Hoe
   B. Spade
   C. Rake
   D. Trowel

23. The energy possessed by a football rolling at a constant speed is
   A. kinetic energy.
   B. potential energy.
   C. electrical energy.
   D. physical energy.

24. Light energy is converted to electrical energy when
   A. electrical oven is in use.
   B. the sun is shining.
   C. solar panels are in use.
   D. cooking on fire.

Turn over

25. One of the properties of solids is that they
A. flow easily.
B. are difficult to compress.
C. have no definite shape.
D. cannot be frozen.

26. Shadow formation shows that light rays
A. can be stopped.
B. travel in straight lines
C. can be bent in any direction
D. move in circles.

27. The storage house of a cell is the
A. nucleus.
B. mitochondrion.
C. vacuole.
D. chloroplast.

28. The main criteria for arranging elements in the periodic table do **not** include
A. shell structure.
B. metals and non-metals.
C. groups and periods.
D. solids and liquids.

29. In which of the following forms of water are particles moving fastest?
A. Cold water
B. Hot water
C. Ice
D. Steam.

30. The region around a magnet where the magnetic force or influence is felt is the
A. polar field.
B. magnetic pressure.
C. magnetic zone.
D. magnetic field.

31. The energy change that occurs in your body when you are running is
A. chemical to mechanical.
B. heat to chemical.
C. chemical to heat.
D. heat to mechanical.

32. The force overcome by a machine is called
A. effort.
B. load.
C. pivot.
D. work

33. Decomposers are organisms that
A. break down dead organisms.
B. produce food.
C. feed on any organism.
D. store energy.

34. The animal used for wool production is
A. cow.
B. sheep.
C. pig.
D. goat.

35. A machine with an efficiency of 60% produces a work output of 300 J. What is the work input?
A. 360 J
B. 400 J
C. 500 J
D. 600 J

36. Which of these food nutrients helps the body grow and repair tissues?
A. Proteins
B. Carbohydrates
C. Vitamins
D. Fats

37. A 3.5kg body is placed on a wall 10m high. Calculate the energy possessed by the body. [Take g = $10ms^{-2}$]
A. 350J
B. 3500J
C. 35.5J
D. 305J

38. In the water cycle, water
I. is continuously recycled.
II. changes from one state to another.
III. moves from one location to another.
IV. created and destroyed.
A. I, II and III only.
B. I, II, III and IV.
C. II, III and IV only.
D. I, III and IV only.

39. Houseflies lay their eggs
A. on food substances.
B. in the soil.
C. in water.
D. on plants.

40. One primary purpose of crop rotation is to
A. increase soil erosion.
B. deplete soil resources.
C. control pests and diseases.
D. control crop yield.

END OF PAPER

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